

Multi-View Representation Learning of Multi-Class Instance Segmentation for Brain Tumors in MRI Imaging

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Abstract—Accurate segmentation of brain tumors from magnetic resonance imaging (MRI) is essential for reliable diagnosis and treatment planning in neuro-oncology. This study proposes a one-stage deep transfer learning-driven framework for multi-class instance segmentation and classification of brain tumors using multi-view MRI imaging, incorporating axial, coronal, and sagittal perspectives. A publicly available dataset consisting of approximately 5,000 grayscale MRI slices was utilized, covering three primary tumor types—Glioma, Meningioma, and Pituitary tumors—with non-tumor images implicitly treated as background. The dataset was manually annotated, preprocessed, and augmented to improve model generalization, yielding a total of 11,581 images for training, validation, and testing. The proposed model was trained using pretrained segmentation weights and evaluated using standard detection and segmentation metrics. Experimental results demonstrate strong performance, achieving an overall segmentation mean Average Precision (mAP) exceeding 0.85 at an IoU threshold of 0.5 and approximately 0.54 at IoU 0.5–0.95. Precision–recall and F1–confidence analyses reveal near-perfect performance for meningioma tumors, balanced detection for pituitary tumors, and robust yet more challenging segmentation for gliomas due to their heterogeneous and infiltrative morphology. Qualitative evaluations across multiple MRI perspectives further confirm accurate tumor localization and anatomically consistent segmentation.

Keywords—brain tumor segmentation, instance segmentation, multi-view mri imaging, medical image analysis, yolov12.

I. INTRODUCTION

Magnetic Resonance Imaging (MRI) remains one of the most reliable and widely adopted imaging modalities for the diagnosis, monitoring, and treatment planning of brain tumors due to its superior soft-tissue contrast and non-invasive nature [1]. By capturing fine-grained anatomical structures across multiple planes and depths, MRI enables clinicians to distinguish pathological tissue from healthy brain matter with high precision [2]. In this study, a publicly available MRI brain tumor dataset compiled in [3] is utilized, consisting of approximately 3,800 grayscale MRI slices acquired at varying depths of the brain. The dataset contains four categories: Glioma (~1,300 images), Meningioma (~1,250 images), Pituitary tumor (~1,250 images), and No tumor (~1,200 images) ensuring a relatively balanced class distribution. Standardized preprocessing techniques, including Gaussian noise reduction, pixel-level annotation, and semantic

segmentation masks, were applied by the dataset authors. The images were further organized into training, validation, and test subsets, providing a consistent experimental foundation. Leveraging these annotations, the present study focuses on employing a one-stage deep transfer learning framework based on YOLOv12 [4] to perform instance-level brain tumor segmentation and multi-class classification directly from MRI data.

Brain tumors originate from abnormal cellular proliferation within the brain or its surrounding structures and can be broadly categorized as benign or malignant depending on their growth behavior, invasiveness, and metastatic potential [5]. Tumors such as meningiomas arise from the meninges and exert pressure on adjacent neural tissues, while gliomas originate from glial cells and are often associated with highly aggressive and infiltrative growth patterns. Pituitary tumors, although frequently benign, can significantly disrupt hormonal regulation and neurological function [6]. Known risk factors include genetic predisposition, family history, prolonged exposure to carcinogens, ionizing radiation, and lifestyle-related contributors such as long-term smoking. In the Philippine context, malignant brain tumors account for approximately 2,480 reported deaths, highlighting the critical need for timely and accurate diagnosis [7]. Manual interpretation of MRI scans by radiologists, while effective, is labor-intensive and subject to inter-observer variability, motivating the integration of automated, AI-driven diagnostic support systems to enhance accuracy, consistency, and clinical throughput.

In recent years, deep learning particularly convolutional neural networks (CNNs) has emerged as a dominant paradigm for medical image analysis [8]. Early studies primarily addressed brain tumor detection as a classification problem. For instance, projection-based classification methods such as those presented in [9] utilized voxel-wise separation strategies combined with softmax classifiers to identify tumor categories. Other approaches employed handcrafted feature extraction followed by classical machine learning classifiers, though these methods often struggled with robustness and generalization across diverse imaging conditions. With the advent of deeper CNN architectures, studies began leveraging models such as AlexNet, VGGNet, and ResNet for MRI-based tumor classification [10], demonstrating notable

improvements in accuracy but limited capability in precise tumor localization.

To overcome these limitations, subsequent research shifted toward segmentation-focused frameworks. Encoder-decoder architectures, particularly U-Net and its variants, have become widely adopted due to their ability to preserve spatial resolution while capturing high-level semantic features. As demonstrated in U-Net-based models achieved high intersection-over-union (IoU) scores even without extensive data augmentation, making them suitable for medical imaging scenarios with limited annotated data. Extensions such as U-Net++, Attention U-Net, and Dense U-Net further improved segmentation accuracy by incorporating skip connections, attention mechanisms, and dense feature reuse. However, these architectures typically operate as multi-stage pipelines and are computationally intensive, limiting their suitability for real-time or resource-constrained environments.

Parallel to encoder-decoder developments, region-based detection frameworks such as Faster R-CNN and Mask R-CNN have also been explored for brain tumor analysis. These models utilize region proposal networks (RPNs) to generate candidate tumor regions before classification and segmentation, enabling precise localization at the cost of increased computational complexity. Studies such as [11] enhanced CNN-based segmentation by incorporating metaheuristic optimization strategies, including the BAT algorithm, to improve convergence and boundary delineation. While these approaches achieved competitive performance, their reliance on multi-stage processing and extensive post-processing remains a key drawback in clinical deployment scenarios.

More recently, there has been growing interest in unified, one-stage deep learning frameworks capable of jointly performing detection, segmentation, and classification in a single forward pass. One-stage models reduce inference latency, simplify training pipelines, and improve computational efficiency, making them attractive for real-time medical applications [12]-[14]. The You Only Look Once (YOLO) family of models has demonstrated remarkable success in object detection and segmentation across domains such as autonomous driving, agriculture, surveillance, manufacturing, and healthcare. Earlier variants, including YOLOv7, YOLOv8, and YOLOv9, introduced architectural refinements that improved accuracy and inference speed, yet their adoption in high-precision medical imaging tasks remains relatively underexplored.

YOLOv12 represents the latest advancement in this family, introducing architectural innovations specifically designed to enhance feature representation and optimization [15]-[16]. These include area attention mechanisms that divide feature maps into equally weighted critical regions, enabling improved spatial awareness and contextual reasoning. The integration of Residual Efficient Layer Aggregation Networks (R-ELAN) [17] enhances gradient flow and scalability, while FlashAttention facilitates high-resolution real-time inference with reduced memory consumption. Additional improvements such as the removal of explicit positional encoding, optimized convolutional restructuring, reduced stacked blocks, and the adoption of a 7×7 convolutional kernel enable more effective encoding of spatial relationships and improved feature abstraction in deeper layers. Collectively, these enhancements address limitations observed in earlier YOLO variants and position

YOLOv12 [18] as a promising candidate for dense prediction tasks such as medical image segmentation.

The objectives of this study are threefold. First, it aims to investigate the applicability of a one-stage deep transfer learning framework based on YOLOv12 for instance-level segmentation and multi-class classification of brain tumors from MRI images. Second, it seeks to quantitatively evaluate the model's detection and segmentation performance using established metrics on a preprocessed and augmented medical dataset. Third, this work contributes to the growing literature on YOLOv12 by demonstrating its effectiveness in a neuro-oncological context, where accurate tumor boundary delineation and robust class discrimination are essential.

This study is delimited to instance segmentation of three common brain tumor types: Glioma, Meningioma, and Pituitary tumors. Although the original dataset includes a No tumor class, these images are excluded from segmentation experiments due to the absence of defined regions of interest and are implicitly treated as background during training. All MRI inputs are grayscale, and each image is assumed to contain a single tumor type, framing the task as a single-instance, multi-class segmentation problem. The experiments, including model training, evaluation, and visualization of learning dynamics, were conducted in Makati City, Metro Manila, using locally available computational resources.

The remainder of this paper is organized as follows. Section II (Methodology) presents the dataset preparation procedures, network architecture, training configuration, and evaluation metrics. Section III (Results and Discussion) provides a detailed quantitative and qualitative analysis of the model's segmentation and classification performance. Finally, Section IV (Conclusion) summarizes the key findings, discusses their implications for medical imaging research, and outlines potential directions for future work.

II. METHODOLOGY

A. Data Preparation, Annotation and Augmentation

The data preparation pipeline commenced with the identification of a suitable medical imaging dataset capable of supporting semantic segmentation and multi-class brain tumor classification. Multiple public repositories: including Google Dataset Search, Kaggle, Roboflow Universe, and ImageNet were surveyed to ensure dataset relevance, quality, and annotation feasibility. Ultimately, a publicly available MRI brain tumor dataset sourced via Google Dataset Search was selected due to its robustness, clear class definitions, and applicability to medical imaging research. The dataset comprises approximately 5,000 grayscale MRI slices categorized into four classes: No Tumor (1,200 images), Pituitary Tumor (1,250 images), Glioma Tumor (1,300 images), and Meningioma Tumor (1,250 images), as illustrated in Fig. 1. The dataset was downloaded in compressed format and imported into Roboflow for annotation and preprocessing. Roboflow was configured using an instance segmentation workspace to enable pixel-level tumor delineation, providing flexibility for future multi-instance scenarios despite the present study focusing on single-tumor images. Manual annotation was performed collaboratively by the research team, with each member assigned a specific tumor class to ensure annotation consistency and workload balance. Following annotation, the dataset was carefully validated to confirm correct mask placement and class labeling before export. The annotated

dataset was then partitioned into training, validation, and testing subsets using a 75–15–15 split, resulting in 8,093 training images and 1,734 images each for validation and testing. Preprocessing steps applied through Roboflow included automatic orientation correction and image resizing to a fixed resolution of 640×640 pixels using stretch normalization. To enhance model generalization and mitigate overfitting, data augmentation techniques were employed, including horizontal flipping, clockwise and counterclockwise 90-degree rotations, and brightness adjustments within a -20% to $+20\%$ range. Each original image was augmented to generate three variants, expanding the dataset to a total of 11,581 images.

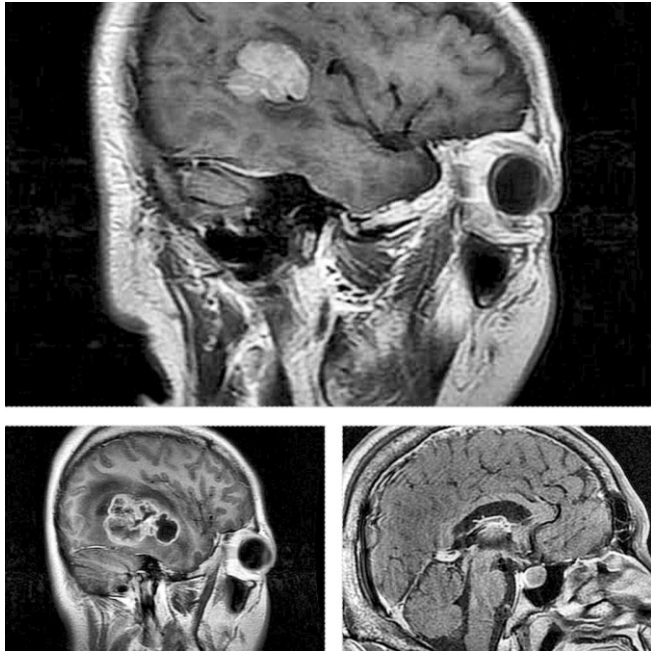


Fig. 1. Meningioma (top), Glioma (bottom left), and Pituitary tumors (bottom right).

An exploratory analysis of the annotated instances revealed a relatively balanced class distribution, with tumor regions predominantly centered within the image frame and exhibiting small-to-medium spatial extents. These characteristics emphasize the need for a segmentation model capable of robust multi-scale feature learning and precise boundary delineation as shown in Fig. 2.

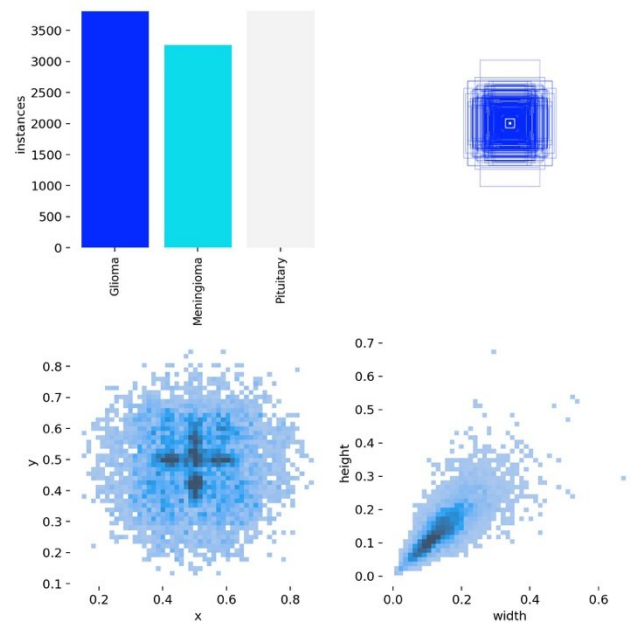


Fig. 2. Exploratory analysis of annotated brain tumor.

B. Model Training and Architecture

Model training was conducted using Google Colab Premium to leverage cloud-based GPU acceleration and reduce computational time, with the training pipeline implemented in Python. The experimental framework was based on a preconfigured training script developed for prior object detection research, which was adapted in this study for medical image segmentation. The environment was initialized with essential deep learning libraries, including PyTorch, Ultralytics, time, augmentations, and associated dependencies, with version control enforced to ensure compatibility and stability throughout training. The YOLOv12 architecture was cloned from its official repository and integrated within a structured project directory hosted on Google Drive, enabling seamless data access and checkpoint storage. Architecturally, YOLOv12 follows a one-stage design composed of three principal components: a backbone, a neck, and a prediction head. The backbone employs a Residual Efficient Layer Aggregation Network (R-ELAN) with enhanced residual connectivity and 7×7 separable convolutions to extract multi-scale spatial features while maintaining computational efficiency. Feature aggregation is further refined within the neck through an area-attention mechanism accelerated by Flash Attention, allowing the model to emphasize critical spatial regions and improve contextual awareness in complex medical imagery. The head transforms these refined feature representations into segmentation masks and class predictions via optimized multi-scale detection pathways. For training, the YOLOv12 segmentation variant was initialized using pretrained segmentation weights to enable effective transfer learning. Hyperparameters were empirically selected, with the maximum number of epochs set to 1000 and early stopping activated using a patience value of 15 epochs to prevent overfitting Fig 3. A batch size of 8 and an input resolution of 640×640 pixels were used, consistent with the preprocessing configuration. Data augmentation parameters included a scaling factor of 0.5, mosaic augmentation probability of 0.5, copy-paste probability of 0.1, and mosaic deactivation 10 epochs prior to early stopping, while mixup augmentation was

disabled. Training was executed on a GPU device, and model checkpoints corresponding to the best-performing and final weights were automatically saved for subsequent evaluation.

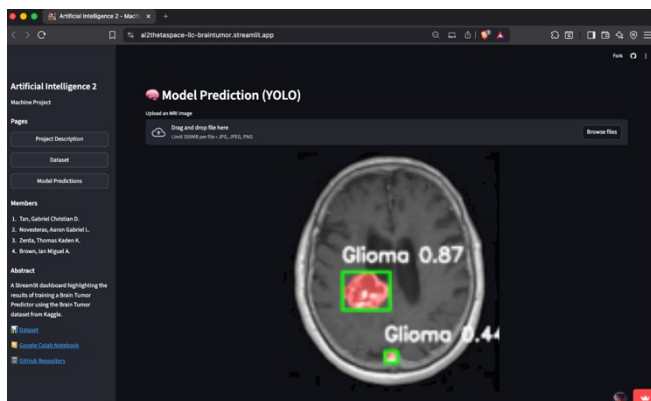


Fig. 3. System Implementation and Streamlit Deployment.

C. Model to Web Deployment

The trained model was deployed using a Streamlit-based web application to demonstrate the complete inference workflow of the proposed brain tumor segmentation and classification system. Streamlit was utilized as the deployment

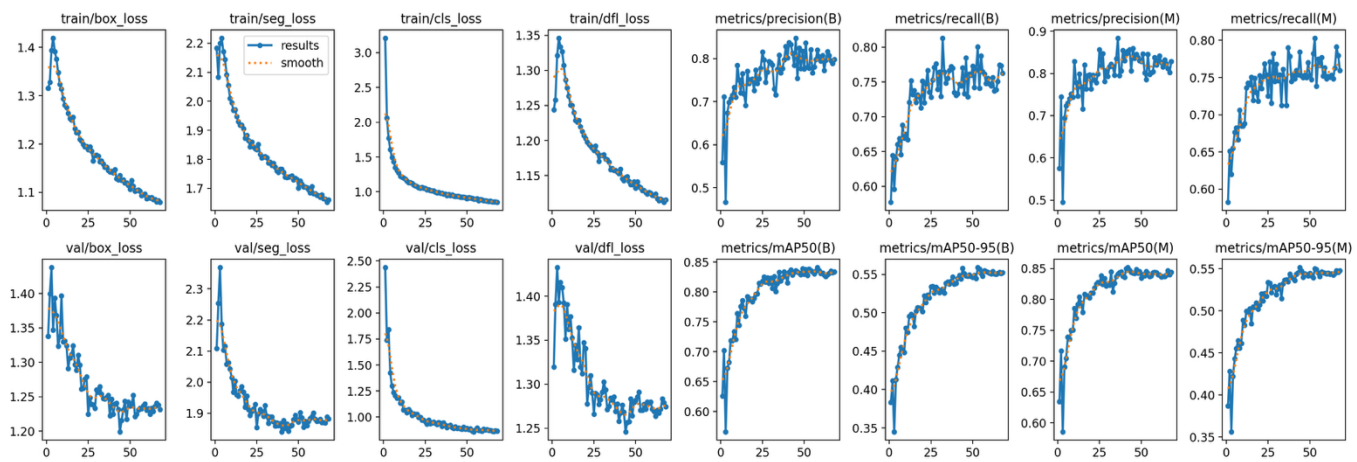


Fig. 4. Training Results.

framework to integrate image preprocessing, model loading, and real-time prediction within an interactive and user-friendly interface. The application allows users to upload MRI images and automatically generates tumor classification and segmentation outputs using the trained model. This deployment setup ensures seamless interaction between the preprocessing pipeline, trained weights, and inference engine, validating the system's operational consistency on unseen multi-view MRI inputs.

III. RESULTS AND DISCUSSIONS

A. Training Results

The training dynamics illustrated in Fig. 3 provide a clear quantitative assessment of the model's learning behavior and segmentation performance across epochs. The segmentation mean Average Precision at IoU 0.5 ($mAP@0.5$) demonstrates a rapid increase during the early training phase, indicating effective feature transfer from the pretrained weights and strong initial convergence. Specifically, $mAP@0.5$ rises sharply within the first 10–15 epochs before gradually stabilizing, ultimately reaching values above 0.85. This trend suggests that the model quickly learns discriminative tumor

features and spatial representations from the MRI data. In contrast, the more stringent $mAP@0.5-0.95$ metric exhibits a slower but steady upward trajectory, converging at approximately 0.54. This behavior reflects the increased difficulty of achieving precise boundary alignment across multiple IoU thresholds, a common challenge in medical image segmentation where tumor margins are often diffuse or irregular.

The observed plateauing of both metrics in later epochs indicates stable convergence and effective regularization, supported by the early stopping mechanism employed during training. The consistent gap between $mAP@0.5$ and $mAP@0.5-0.95$ further highlights the model's robustness in tumor localization while underscoring the inherent complexity of fine-grained boundary delineation in brain MRI scans. Importantly, the absence of significant oscillations or degradation in performance during later epochs suggests that the selected hyperparameters and augmentation strategies successfully mitigated overfitting.

B. Precision–Recall Analysis

Fig. 4 presents the precision–recall (PR) curves for the three tumor classes: Glioma, Meningioma, and Pituitary, along with the aggregated performance across all classes at an

IoU threshold of 0.5. The overall model achieves a mean Average Precision of $mAP@0.5 = 0.846$, indicating strong detection and segmentation capability across tumor types. Among the individual classes, Meningioma demonstrates the highest performance with an AP of 0.982, maintaining near-perfect precision across a wide recall range up to approximately 0.95. This suggests that meningioma tumors exhibit highly distinguishable morphological and intensity patterns in MRI scans, enabling the model to achieve both high confidence and comprehensive coverage with minimal false positives.

The Pituitary tumor class attains an AP of 0.802, reflecting balanced performance between precision and recall. The PR curve shows a gradual decline in precision as recall exceeds approximately 0.80, indicating that while the model successfully detects most pituitary tumors, boundary ambiguity and size variability may introduce moderate false positives at higher recall levels. Similarly, the Glioma class records the lowest performance with an AP of 0.754, characterized by a more pronounced drop in precision beyond a recall of approximately 0.70. This behavior highlights the intrinsic difficulty of glioma segmentation, as gliomas often present diffuse, infiltrative boundaries and heterogeneous

textures that complicate precise localization and increase misclassification rates Fig 5.

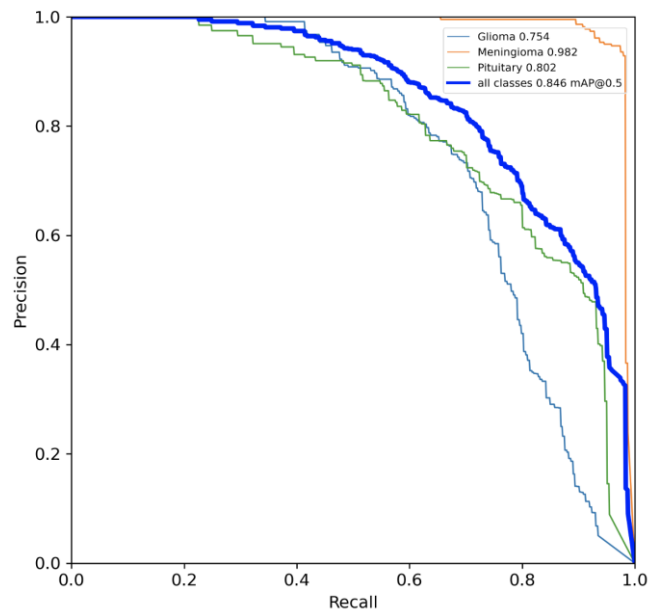


Fig. 5. Precision-Recall (PR) Curves.

Across all classes, the aggregated PR curve exhibits a smooth and stable trade-off between precision and recall, with precision remaining above 0.80 until recall approaches approximately 0.75. The convergence of all curves toward low precision at near-perfect recall reflects the expected increase in false positives when the detection threshold is relaxed.

C. F1-Confidence Curve Analysis

Fig. 6 illustrates the relationship between the F1 score and the confidence threshold for each tumor class, as well as the aggregated performance across all classes. The overall F1 score for all classes peaks at approximately 0.80 when the confidence threshold is set to 0.304, indicating an optimal balance between precision and recall at this operating point. Below this threshold, the inclusion of low-confidence predictions increases recall but introduces a higher proportion of false positives, resulting in reduced F1 scores. Conversely, thresholds above this value lead to a rapid decline in recall, particularly beyond confidence levels of 0.6–0.7, thereby diminishing overall performance.

Class-wise analysis reveals notable performance disparities among tumor types. Meningioma achieves the highest and most stable F1 performance, maintaining an F1 score above 0.90 across a broad confidence range from approximately 0.05 to 0.70. This stability indicates strong class separability and high confidence predictions, consistent with the class's near-perfect mAP@0.5 observed in earlier analyses. In contrast, Glioma and Pituitary tumors exhibit lower peak F1 scores, reaching maxima of approximately 0.71 and 0.72, respectively, near confidence thresholds of 0.30–0.35. Beyond a confidence level of approximately 0.60, both classes experience a pronounced drop in F1, reflecting increased false negatives as detection thresholds become more restrictive.

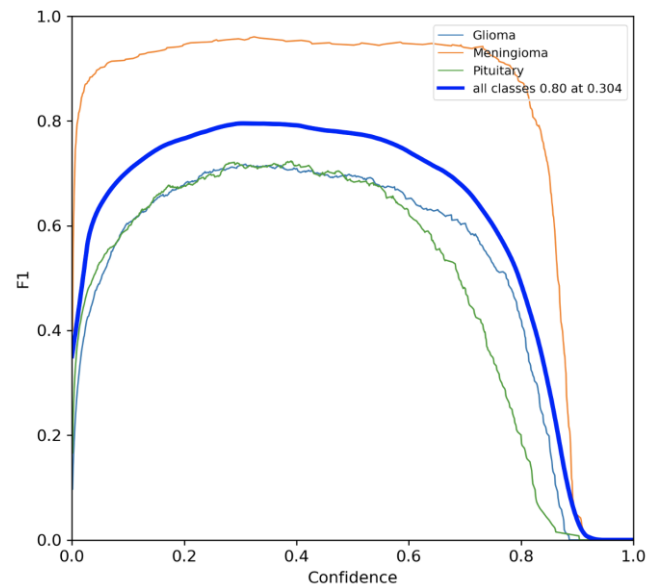


Fig. 6. F1-Confidence Curves.

The sharper decline observed for the Pituitary class suggests heightened sensitivity to confidence threshold selection, likely attributable to variability in tumor size, intensity contrast, and anatomical location within MRI scans. The Glioma curve exhibits a similar but more gradual degradation, consistent with the infiltrative and heterogeneous nature of gliomas, which complicates consistent boundary prediction.

D. Qualitative Evaluation of Segmentation Results

The qualitative results presented in Fig. 7 demonstrate the model's ability to accurately localize and segment brain tumors across multiple MRI slices and anatomical planes. For pituitary tumors, the model consistently identifies centrally located lesions within the sellar region, producing compact and well-aligned bounding boxes and segmentation masks that closely conform to the ground truth. The predicted regions exhibit minimal background leakage and stable spatial consistency across axial and coronal views, indicating strong sensitivity to the characteristic intensity and shape patterns of pituitary tumors. In contrast, glioma predictions exhibit greater variability in both size and morphology, reflecting the infiltrative and heterogeneous nature of gliomas. Despite these challenges, the model successfully captures irregular tumor boundaries and extended regions of abnormal tissue, with segmentation masks adapting to non-uniform shapes and asymmetric growth patterns. Some predictions show partial under-segmentation at diffuse margins, which aligns with the lower quantitative performance observed for gliomas in the precision-recall and F1 analyses.

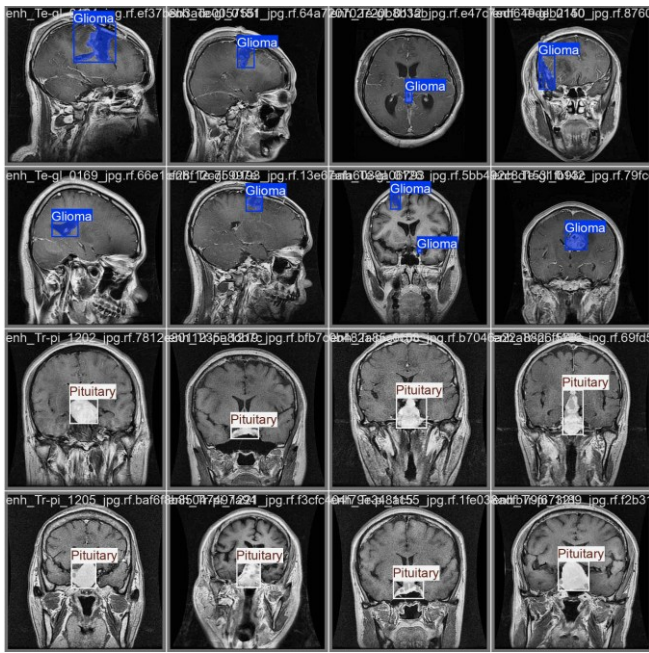


Fig. 7. Qualitative Results.

IV. CONCLUSIONS

This study successfully demonstrated the effectiveness of a one-stage deep transfer learning framework for multi-class instance segmentation and classification of brain tumors from MRI images. Leveraging a YOLOv12-based architecture, the proposed approach achieved strong quantitative performance, with an overall segmentation mAP@0.5 exceeding 0.85 and mAP@0.5–0.95 reaching approximately 0.54, indicating robust tumor localization and competitive boundary delineation across varying IoU thresholds. Class-wise analyses further revealed superior performance for meningioma tumors, achieving near-perfect precision–recall characteristics and consistently high F1 scores, while pituitary tumors exhibited balanced detection performance with moderate sensitivity to confidence threshold selection. Glioma segmentation, although more challenging due to infiltrative and heterogeneous tumor morphology, maintained reliable localization and acceptable precision, as reflected in both quantitative metrics and qualitative visual assessments. Exploratory dataset analysis confirmed balanced class representation and centralized tumor spatial distributions, supporting stable model convergence. Training dynamics showed rapid early learning followed by stable convergence without overfitting, and optimal operating points were identified through F1–confidence analysis, with a global confidence threshold of approximately 0.30 yielding the best balance between precision and recall. Qualitative evaluation further validated the model’s capability to produce anatomically consistent and clinically interpretable segmentation outputs across multiple MRI views. Collectively, these findings highlight the potential of one-stage, attention-enhanced deep learning frameworks for efficient and accurate MRI-based brain tumor analysis, offering a promising foundation for future clinical decision-support systems and further research involving larger datasets, multi-instance scenarios, and cross-institutional validation.

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